



HOSPITAL MANAGEMENT SYSTEM

Lab Report

Subject: Object Oriented Programming (OOP)

Technology: Java 25 | GUI: JavaFX | Database: MySQL

Platform: Kali Linux | IDE: IntelliJ IDEA Ultimate

Submitted By:

Roll No: _____

Semester: _____

Program: _____

Department: _____

Submitted To:

1. Introduction

This lab report documents the complete design, development, and implementation of a Hospital Management System (HMS) built using Java with JavaFX for the Graphical User Interface and MySQL as the backend database. The system automates and manages key hospital operations including patient registration, room management, medical records, medicine dispensary, billing, report generation, and secure user access control with Email-based Two-Factor Authentication (2FA).

The project demonstrates core Object-Oriented Programming (OOP) concepts such as inheritance, encapsulation, polymorphism, and abstraction, alongside real-world software engineering practices including role-based access control, session management, database integration, and enterprise-level security features.

2. Objectives

- Develop a fully functional Hospital Management System with JavaFX GUI
- Implement role-based access control for Clerk, Physician, and Nurse
- Integrate MySQL database for persistent data storage with full CRUD operations
- Apply OOP principles (Inheritance, Encapsulation, Polymorphism, Abstraction) throughout
- Implement comprehensive security: login attempts, account locking, session timeout, password policies
- Add Email Two-Factor Authentication (2FA) using Gmail SMTP
- Create Medicine/Dispensary management module with prescription tracking
- Implement Bill Generation with PDF export and Paid/Unpaid tracking
- Generate PDF and CSV reports for patients, prescriptions, rooms, and activity logs
- Maintain complete audit trails via activity logs

3. Technology Stack

Component	Technology	Version/Detail
Programming Language	Java (OpenJDK)	25.0.3
GUI Framework	JavaFX SDK	21 & 25

Database	MySQL	Latest
IDE	IntelliJ IDEA	Ultimate 2026
Operating System	Kali Linux	Latest
DB Connector	MySQL Connector/J	9.3.0
Email Library	Jakarta Mail	2.0.1
PDF Library	iText PDF	5.5.13.3
Activation Library	Jakarta Activation API	2.1.3

4. System Architecture

The Hospital Management System follows a layered architecture pattern with clear separation of concerns across five distinct layers:

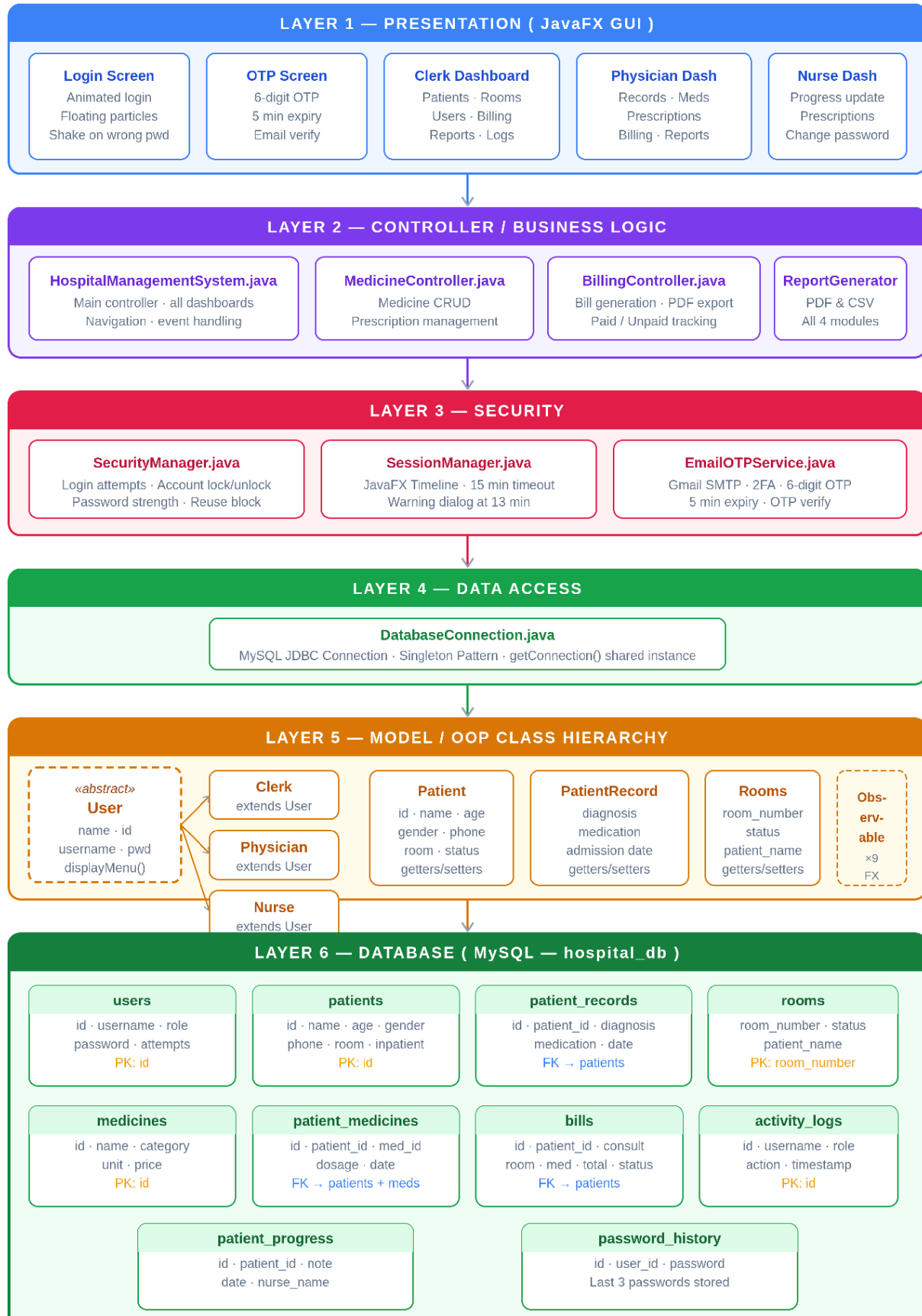
Layer	Components	Responsibility
Presentation Layer	JavaFX (Scenes, Tabs, Tables)	GUI, user interaction, visual feedback
Controller Layer	HospitalManagementSystem.java, MedicineController.java, BillingController.java	Business logic, event handling, UI coordination
Security Layer	SecurityManager.java, SessionManager.java, EmailOTPService.java	Auth, 2FA OTP, session timeout, password policies
Report Layer	ReportGenerator.java	PDF & CSV report generation for all modules
Data Layer	DatabaseConnection.java	MySQL JDBC connection & query execution
Model Layer	Patient, User, Rooms, PatientRecord (Roles.*)	OOP data models & class hierarchy

5. Architectural Diagram

The following diagram illustrates the complete system architecture of the Hospital Management System, showing the relationships between all layers, components, and the database:

HOSPITAL MANAGEMENT SYSTEM

SYSTEM ARCHITECTURE DIAGRAM

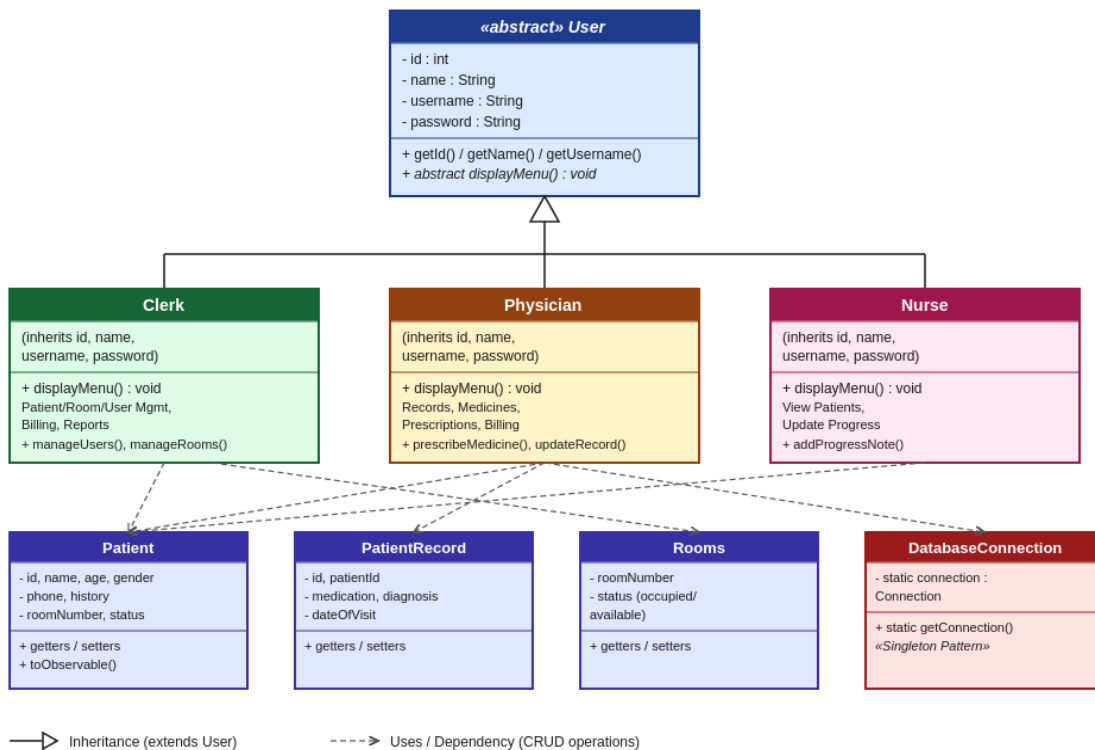


5.1 Component Interaction Flow

From	To	Interaction
Login Screen	SecurityManager	Check lock status, record attempts
Login Screen	EmailOTPService	Generate & send OTP via Gmail SMTP
OTP Screen	EmailOTPService	Verify OTP, check expiry
All Dashboards	SessionManager	Reset 15-min timer on activity, Warning On 13 min.
Clerk Dashboard	DatabaseConnection	Patient CRUD, Room, User management
Physician Dashboard	MedicineController	Medicine & prescription management
Clerk/Physician	BillingController	Generate bills, export PDF
Clerk/Physician	ReportGenerator	Export PDF/CSV reports
All Actions	activity_logs table	Log every user action with timestamp

6. Class Diagram

Class Diagram - Hospital Management System (OOP Model)



7. OOP Concepts Applied

7.1 Inheritance

The User class serves as an abstract base class. Clerk, Physician, and Nurse all extend User, inheriting common attributes and implementing their own displayMenu() method.

Figure: Class Diagram showing the abstract User base class, its derived role classes (Clerk, Physician, Nurse), and associated model/data classes.

Base Class	Derived Classes	Inherited Attributes
User (Abstract)	Clerk, Physician, Nurse	name, id, username, password
User	Clerk	displayMenu() — Add/View/Admit/Discharge Patients, User Management
User	Physician	displayMenu() — Records, Medicines, Prescriptions, Billing
User	Nurse	displayMenu() — View Patients, Update Progress

7.2 Encapsulation

All class attributes are declared private with public getter and setter methods. DatabaseConnection uses a private static connection with a public static getConnection() method — classic Singleton pattern with encapsulation.

7.3 Polymorphism

The showDashboard(User user) method demonstrates runtime polymorphism — it accepts a User object but dynamically calls showClerkDashboard(), showPhysicianDashboard(), or showNurseDashboard() based on the actual object type at runtime using instanceof checks.

7.4 Abstraction

The User class is declared abstract with an abstract method displayMenu(). This forces all subclasses to provide their own implementation, hiding implementation details and exposing only what is necessary.

8. Database Design



8.1 Tables Explanation

The database hospital_db consists of 10 tables designed with proper relational integrity, foreign key constraints, and ON DELETE CASCADE rules:

Table Name	Primary Key	Purpose
users	id (AUTO_INCREMENT)	Store Clerk, Physician, Nurse accounts with lock/attempt tracking
patients	id (AUTO_INCREMENT)	Patient personal & medical information
patient_records	id (AUTO_INCREMENT)	Admission records — medication & diagnosis
patient_progress	id (AUTO_INCREMENT)	Daily progress notes by nurses/physicians
rooms	room_number	10 hospital rooms — occupied/available status
medicines	id (AUTO_INCREMENT)	Medicine list with category, unit, price
patient_medicines	id (AUTO_INCREMENT)	Prescriptions — patient to medicine mapping
bills	id (AUTO_INCREMENT)	Patient bills — consultation, room, medicine charges
activity_logs	id (AUTO_INCREMENT)	Complete audit trail of all user actions
password_history	id (AUTO_INCREMENT)	Last 3 passwords to prevent reuse

8.2 Key Relationships

- patients -> patient_records (1:Many) — One patient, multiple records
- patients -> patient_progress (1:Many) — One patient, multiple progress entries
- patients -> patient_medicines (1:Many) — One patient, multiple prescriptions
- patients -> bills (1:Many) — One patient, multiple bills
- medicines -> patient_medicines (1:Many) — One medicine, multiple prescriptions
- patients -> rooms (1:1) — One room per patient at a time
- users -> password_history (1:Many) — Track last 3 passwords per user



9. Roles & Features

9.1 Clerk

Feature	Description
Patient Management	Add, Edit, Delete, View patients with real-time search/filter
Admit Patient	Assign room, add medication and date of visit
Discharge Patient	Release patient from room, update inpatient status
Room Management	View room status (color coded), release occupied rooms
User Management	Add, Edit, Delete, Lock/Unlock users for all roles
Billing	Generate bills (consultation + room + medicine), mark Paid/Unpaid, export PDF
Reports	Generate PDF & CSV for patients, prescriptions, rooms, activity logs
Patient Portal	Search by name/ID and view full patient details, records & progress
Activity Logs	View and clear all system activity logs
Change Password	Secure password change with strength validation & reuse block

9.2 Physician

Feature	Description
Patient View	View, search, and filter all patients
Medical Records	Add and update medication and diagnosis per patient
Medicine Management	Add, Edit, Delete medicines with category, unit, price
Prescribe Medicine	Assign medicines with dosage to patients, remove prescriptions
Billing	Generate and manage patient bills, export PDF invoices
Reports	Generate PDF & CSV reports for all modules
Patient Portal	Full patient details, records & progress history view
Activity Logs	View complete system activity logs
Change Password	Secure password change

9.3 Nurse

Feature	Description
Patient View	View and search all patients
Progress Update	Add daily progress notes with date and nurse name
Progress History	View complete progress history per patient
Prescriptions View	View all patient prescriptions
Change Password	Secure password change

10. Security Features

Security Feature	Implementation	Effect
Email 2FA (OTP)	EmailOTPService.java + Gmail SMTP	6-digit OTP after password — 5 min expiry
Login Attempt Limit	Max 3 failed attempts tracked in DB	Account auto-locks after 3 failures
Account Lock/Unlock	Clerk can lock/unlock any account	Prevents unauthorized access
Session Timeout	SessionManager with JavaFX Timeline	Auto logout after 15 min inactivity
Session Warning	Warning dialog at 13 minutes	Alerts user 2 min before auto logout
Password Strength	SecurityManager.checkPassword Strength()	Min 8 chars + uppercase + number + special char
Password Reuse Block	Last 3 passwords stored in DB	Cannot reuse last 3 passwords
Activity Logging	All actions logged with timestamp	Full audit trail for accountability
Failed Login Tracking	Logged in activity_logs table	Security monitoring & forensics
Animated Login	JavaFX animations + floating particles	Visual security feedback (shake on wrong password)

11. Medicine / Dispensary Module

Feature	Who	Details
Add Medicine	Physician	Name, Category, Unit, Price — with search
Edit Medicine	Physician	Update any medicine detail
Delete Medicine	Physician	Remove medicine from dispensary
Search Medicine	All	Real-time search by name or category
Prescribe Medicine	Physician	Assign medicine + dosage to patient with date
Remove Prescription	Physician	Remove a patient prescription
View All Prescriptions	Clerk, Physician, Nurse	Complete prescription history with search

12. Billing Module

Feature	Details
Generate Bill	Select patient, enter consultation fee, room charges x days, medicine charges
Auto Total	Total calculated automatically as values are entered
Status Tracking	Mark bill as Paid or Unpaid
PDF Invoice	Professional PDF bill with hospital header, breakdown, total, status
View All Bills	Complete bill history with all charges and status
Mark as Paid	Update bill status from Unpaid to Paid
Delete Bill	Remove bill with confirmation dialog
Print/Re-export	Re-generate PDF for any existing bill

13. Reports Module

Report Type	PDF	CSV	Contents
Patient Report	Yes	Yes	ID, Name, Age, Gender, Phone, History, Room, Status
Prescription Report	Yes	Yes	Patient, Medicine, Category, Dosage, Prescribed By, Date
Room Occupancy Report	Yes	Yes	Room No, Status (color coded), Patient Name
Activity Logs Report	Yes	Yes	Time, Username, Role, Action

14. Project File Structure

File	Type	Purpose
HospitalManagementSystem.java	Main Class	JavaFX app, animated login, all dashboards & UI
Database/DatabaseConnection.java	Database	MySQL JDBC connection (Singleton pattern)
Security/SecurityManager.java	Security	Login attempts, lock, password strength & history
Security/SessionManager.java	Security	Auto logout 15 min, warning dialog at 13 min
Security/EmailOTPService.java	Security	Gmail SMTP 2FA — OTP generate, send, verify
Controller/MedicineController.java	Controller	Medicine CRUD & prescription management
Controller/BillingController.java	Controller	Bill generation, PDF invoice, Paid/Unpaid
Controller/ReportGenerator.java	Reports	PDF & CSV reports for all 4 report types
Models/ObservablePatient.java	Model	JavaFX observable for patient TableView
Models/ObservableRoom.java	Model	JavaFX observable for room TableView
Models/ObservableRecord.java	Model	JavaFX observable for records TableView
Models/ObservableProgress.java	Model	JavaFX observable for progress TableView

Models/ObservableUser.java	Model	JavaFX observable for user management TableView
Models/ObservableLog.java	Model	JavaFX observable for activity logs TableView
Models/ObservableMedicine.java	Model	JavaFX observable for medicine TableView
Models/ObservablePrescription.java	Model	JavaFX observable for prescriptions TableView
Models/ObservableBill.java	Model	JavaFX observable for billing TableView
Roles/User.java	OOP Base	Abstract base class for all user roles
Roles/Clerk.java	OOP Model	Clerk role extending User
Roles/Physician.java	OOP Model	Physician role extending User
Roles/Nurse.java	OOP Model	Nurse role extending User
Roles/Patient.java	OOP Model	Patient data model with getters/setters
Roles/PatientRecord.java	OOP Model	Patient record data model
Roles/Rooms.java	OOP Model	Room data model

15. Default Login Credentials

Role	Username	Password	Access Level
Clerk	clerk	123	Full patient, user, billing & report management
Physician	physician	456	Medical records, medicines, billing & reports
Nurse	nurse	789	Patient progress & prescription view

16. Testing

Test Case	Input	Expected Output	Result
Valid Login	clerk / 123	OTP sent to email	Pass
OTP Verification	Correct 6-digit OTP	Clerk dashboard opens	Pass
Wrong OTP	Incorrect OTP	Error message shown	Pass
Expired OTP	OTP after 5 min	Expired error shown	Pass
Invalid Login	clerk / wrongpass	Error + attempts shown	Pass
Account Lock	3 wrong passwords	Account locked message	Pass
Unlock Account	Clerk unlocks user	User can login again	Pass
Add Patient	Valid patient data	Patient saved in DB	Pass
Edit Patient	Select + update	Patient updated in DB	Pass
Delete Patient	Select + delete	Patient removed from DB	Pass
Admit Patient	Select room + medication	Room assigned, inpatient=true	Pass
Discharge Patient	Select admitted patient	Room released, inpatient=false	Pass
Add Medicine	Name + Category	Medicine saved in DB	Pass
Prescribe Medicine	Patient + Medicine + Dosage	Prescription saved	Pass
Generate Bill	Fees + charges	Bill saved, total calculated	Pass
Export Bill PDF	Click Generate PDF	PDF file saved to disk	Pass
Mark Bill Paid	Select bill	Status updated to Paid	Pass
Generate Report PDF	Click any report	PDF saved to disk	Pass
Generate Report CSV	Click any CSV	CSV saved to disk	Pass
Auto Logout	15 min inactivity	Redirected to login	Pass
Weak Password	Password: abc	Strength error shown	Pass
Password Reuse	Old password	Reuse error shown	Pass
Search Patient	Type patient name	Filtered results shown	Pass

17. Conclusion

The Hospital Management System successfully demonstrates the application of Object-Oriented Programming principles in a comprehensive real-world project. The system provides a complete solution for hospital operations management with an animated JavaFX graphical interface, robust MySQL database integration, and enterprise-level security features.

Key achievements of this project include:

- Successfully implemented OOP hierarchy with abstract User base class and three concrete role subclasses using Inheritance, Encapsulation, Polymorphism, and Abstraction
- Developed a complete CRUD-based patient management system with admit, discharge, and room management
- Implemented Email Two-Factor Authentication (2FA) using Gmail SMTP for enhanced login security
- Created comprehensive security framework with login attempt limits, account locking, session timeout, and password policies
- Built a medicine dispensary module for managing hospital medicines and patient prescriptions
- Implemented bill generation with automatic total calculation, PDF invoice export, and payment tracking
- Created a complete reports module generating PDF and CSV reports for patients, prescriptions, rooms, and activity logs
- Integrated real-time activity logging providing complete audit trail functionality
- Implemented role-based access control ensuring each user can only perform authorized operations
- Added animated login screen with floating particles, slide-in effects, and smooth transitions

This project provided valuable hands-on experience with Java, JavaFX, MySQL, JDBC, Gmail SMTP, iText PDF generation, and software engineering concepts applicable to enterprise-level application development.

18. References

1. Oracle Java Documentation — <https://docs.oracle.com/en/java/>
2. JavaFX Documentation — <https://openjfx.io>
3. MySQL Documentation — <https://dev.mysql.com/doc/>
4. MySQL Connector/J — <https://dev.mysql.com/doc/connector-j/>
5. Jakarta Mail Documentation — <https://eclipse-ee4j.github.io/mail/>
6. iText PDF Documentation — <https://itextpdf.com/> &
7. IntelliJ IDEA Documentation — <https://www.jetbrains.com/idea/documentation/>
8. Gluon JavaFX SDK — <https://gluonhq.com/products/javafx/>

18.1 Libraries

1. Gluon JavaFX SDK — https://download2.gluonhq.com/openjfx/21.0.2/openjfx-21.0.2_linux-x64_bin-sdk.zip
 2. JDBC Connection — <https://dev.mysql.com/get/Downloads/Connector-J/mysql-connector-j-8.3.0.tar.gz>
 3. Jakarta Mail — <https://repo1.maven.org/maven2/com/sun/mail/jakarta.mail/2.0.1/jakarta.mail-2.0.1.jar>
 4. Jakarta Mail Activation — <https://repo1.maven.org/maven2/jakarta/activation/jakarta.activation-api/2.1.3/jakarta.activation-api-2.1.3.jar>
 5. Itext PDF — <https://repo1.maven.org/maven2/com/itextpdf/itextpdf/5.5.13.3/itextpdf-5.5.13.3.jar>
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